

RedPitaya

FPGA memory map

Written by	Revision Description	Version	Date
Matej Oblak	Initial	0.1	08/11/13
Matej Oblak	Release1 update	0.2	16/12/13
Matej Oblak	ASG - added burst mode ASG - buffer read pointer readout		Dec. 2014 Jan. 2015
Matej Oblak	AXI master documented		Feb. 2015
Iztok Jeras	Added debounce delay register		Mar. 2015
Iztok Jeras	Added digital loopback, pre trigger status		Apr. 2015
Iztok Jeras	Removed XADC registers GPIO[0] is now R/W		Avg. 2015

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About Document

Red Pitaya HDL design has multiple functions, which are configured by registers. It also uses memory locations to store capture data and generate output signals. All of this are described in this document. Memory location is written in a way that is seen by SW.

FPGA Memory Map

The table describes address space partitioning implemented on FPGA via AXI GP0 interface. All registers have offsets aligned to 4 bytes and are 32 -bit wide. Granularity is 32-bit, meaning that minimum transfer size is 4 bytes. The organization is little-endian .

The memory block is divided into 8 parts. Each part is occupied by individual IP core. Address space of individual application is described in the subsection below. The size of each IP core address space is 4MByte.

For additional informations and better understanding check other documents (schematics, specifications...).

	Start	End	Module Name
CS[0]	0x40000000	0x400FFFFFF	Housekeeping
CS[1]	0x40100000	0x401FFFFFF	Oscilloscope
CS[2]	0x40200000	0x402FFFFFF	Arbitrary signal generator (ASG)
CS[3]	0x40300000	0x403FFFFFF	PID controller
CS[4]	0x40400000	0x404FFFFFF	Analog mixed signals (AMS)
CS[5]	0x40500000	0x405FFFFFF	Daisy chain
CS[6]	0x40600000	0x406FFFFFF	FREE
CS[7]	0x40700000	0x407FFFFFF	Power test

Red Pitaya Modules

Here are described submodules used in Red Pitaya FPGA logic.

Housekeeping

offset	description	bits	R/W
0x0	ID		
	<i>Reserved</i>	<i>31:4</i>	<i>R</i>
	Design ID 0-prototype0, 1-release1	3:0	R
0x4	DNA part1		
	DNA[31:0]	31:0	R
0x8	DNA part2		
	<i>Reserved</i>	<i>31:25</i>	<i>R</i>
	DNA[56:32]	24:0	R
0xC	Digital Loopback		
	<i>Reserved</i>	<i>31:1</i>	<i>R</i>
	digital_loop	0	R/W
0x10	Expansion connector direction P		
	<i>Reserved</i>	<i>31:8</i>	<i>R</i>
	Direction for P lines 1-out 0-in	7:0	R/W
0x14	Expansion connector direction N		
	<i>Reserved</i>	<i>31:8</i>	<i>R</i>
	Direction for N lines 1-out 0-in	7:0	R/W
0x18	Expansion connector output P		
	<i>Reserved</i>	<i>31:8</i>	<i>R</i>
	P pins output	7:0	R/W
0x1C	Expansion connector output N		
	<i>Reserved</i>	<i>31:8</i>	<i>R</i>
	N pins output	7:0	R/W
0x20	Expansion connector input P		
	<i>Reserved</i>	<i>31:8</i>	<i>R</i>
	P pins input	7:0	R
0x24	Expansion connector input N		
	<i>Reserved</i>	<i>31:8</i>	<i>R</i>
	N pins input	7:0	R
0x30	LED control		
	<i>Reserved</i>	<i>31:8</i>	<i>R</i>

	LEDs 7-0	7:0	R/W
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Oscilloscope

offset	description	bits	R/W
0x0	Configuration		
	<i>Reserved</i>	31:3	R
	Trigger status before acquire ends (0 – pre trigger, 1 – post trigger)	2	R
	Reset write state machine	1	W
	Start writing data into memory (ARM trigger).	0	W
0x4	Trigger source		
	Selects trigger source for data capture. When trigger delay is ended value goes to 0.		
	<i>Reserved</i>	31:4	R
	Trigger source: 1-trig immediately 2-ch A threshold positive edge 3-ch A threshold negative edge 4-ch B threshold positive edge 5-ch B threshold negative edge 6-external trigger positive edge - DIO0_P pin 7-external trigger negative edge 8-arbitrary wave generator application positive edge 9-arbitrary wave generator application negative edge	3:0	R/W
0x8	Ch A threshold		
	<i>Reserved</i>	31:14	R
	Ch A threshold, makes trigger when ADC value cross this value	13:0	R/W
0xC	Ch B threshold		
	<i>Reserved</i>	31:14	R
	Ch B threshold, makes trigger when ADC value cross this value	13:0	R/W
0x10	Delay after trigger		
	Number of decimated data after trigger written into memory	31:0	R/W
0x14	Data decimation		
	Decimate input data, uses data average		
	<i>Reserved</i>	31:17	R
	Data decimation, supports only this values: 1,8, 64,1024,8192,65536. If other value is written data will NOT be correct.	16:0	R/W
0x18	Write pointer - current		
	<i>Reserved</i>	31:14	R
	Current write pointer	13:0	R
0x1C	Write pointer - trigger		
	<i>Reserved</i>	31:14	R

	Write pointer at time when trigger arrived	13:0	R
0x20	Ch A hysteresis		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	Ch A threshold hysteresis. Value must be outside to enable trigger again.	13:0	R/W
0x24	Ch B hysteresis		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	Ch B threshold hysteresis. Value must be outside to enable trigger again.	13:0	R/W
0x28	Other		
	<i>Reserved</i>	<i>31:1</i>	<i>R</i>
	Enable signal average at decimation	0	R/W
0x2C	PreTrigger Counter		
	This unsigned counter holds the number of samples captured between the start of acquire and trigger. The value does not overflow, instead it stops incrementing at 0xffffffff.	31:0	R
0x30	CH A Equalization filter		
	<i>Reserved</i>	<i>31:18</i>	<i>R</i>
	AA Coefficient	17:0	R/W
0x34	CH A Equalization filter		
	<i>Reserved</i>	<i>31:25</i>	<i>R</i>
	BB Coefficient	24:0	R/W
0x38	CH A Equalization filter		
	<i>Reserved</i>	<i>31:25</i>	<i>R</i>
	KK Coefficient	24:0	R/W
0x3C	CH A Equalization filter		
	<i>Reserved</i>	<i>31:25</i>	<i>R</i>
	PP Coefficient	24:0	R/W
0x40	CH B Equalization filter		
	<i>Reserved</i>	<i>31:18</i>	<i>R</i>
	AA Coefficient	17:0	R/W
0x44	CH B Equalization filter		
	<i>Reserved</i>	<i>31:25</i>	<i>R</i>
	BB Coefficient	24:0	R/W
0x48	CH B Equalization filter		
	<i>Reserved</i>	<i>31:25</i>	<i>R</i>
	KK Coefficient	24:0	R/W
0x4C	CH B Equalization filter		

	<i>Reserved</i>	31:25	R
	PP Coefficient	24:0	R/W
0x50	CH A AXI lower address		
	Starting writing address	31:0	R/W
0x54	CH A AXI upper address		
	Address where it jumps to lower	31:0	R/W
0x58	CH A AXI delay after trigger		
	Number of decimated data after trigger written into memory	31:0	R/W
0x5C	CH A AXI enable master		
	<i>Reserved</i>	31:1	R
	Enable AXI master	0	R/W
0x60	CH A AXI write pointer - trigger		
	Write pointer at time when trigger arrived	31:0	R
0x64	CH A AXI write pointer - current		
	Current write pointer	31:0	R
0x70	CH B AXI lower address		
	Starting writing address	31:0	R/W
0x74	CH B AXI upper address		
	Address where it jumps to lower	31:0	R/W
0x78	CH B AXI delay after trigger		
	Number of decimated data after trigger written into memory	31:0	R/W
0x7C	CH B AXI enable master		
	<i>Reserved</i>	31:1	R
	Enable AXI master	0	R/W
0x80	CH B AXI write pointer - trigger		
	Write pointer at time when trigger arrived	31:0	R
0x84	CH B AXI write pointer - current		
	Current write pointer	31:0	R
0x90	Trigger debouncer time		
	Number of ADC clock periods trigger is disabled after activation reset value is decimal 62500 or equivalent to 0.5ms	19:0	R/W
0xA0	Accumulator data sequence length		
	<i>Reserved</i>	31:14	R
0xA4	Accumulator data offset corection ChA		
	<i>Reserved</i>	31:14	R
	signed offset value	13:0	R/W
0xA8	Accumulator data offset corection ChB		
	<i>Reserved</i>	31:14	R

	signed offset value	13:0	R/W
0x10000 to 0x1FFFC	Memory data (16k samples)		
	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Captured data for ch A	15:0	R
0x20000 to 0x2FFFC	Memory data (16k samples)		
	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Captured data for ch B	15:0	R

Arbitrary Signal Generator (ASG)

offset	description	bits	R/W
0x0	Configuration		
	<i>Reserved</i>	31:25	R
	ch B external gated repetitions	24	R/W
	ch B set output to 0	23	R/W
	ch B SM reset	22	R/W
	<i>Reserved</i>	21	R/W
	ch B SM wrap pointer (if disabled starts at address 0)	20	R/W
	ch B trigger selector: (don't change when SM is active) 1-trig immediately 2-external trigger positive edge - DIO0_P pin 3-external trigger negative edge	19:16	R/W
	<i>Reserved</i>	15:9	R
	ch A external gated bursts	8	R/W
	ch A set output to 0	7	R/W
	ch A SM reset	6	R/W
	<i>Reserved</i>	5	R/W
	ch A SM wrap pointer (if disabled starts at address 0)	4	R/W
	ch A trigger selector: (don't change when SM is active) 1-trig immediately 2-external trigger positive edge - DIO0_P pin 3-external trigger negative edge	3:0	R/W
0x4	Ch A amplitude scale and offset		
	out = (data*scale)/0x2000 + offset		
	<i>Reserved</i>	31:30	R
	Amplitude offset	29:16	R/W
	<i>Reserved</i>	15:14	R
	Amplitude scale. 0x2000 == multiply by 1. Unsigned	13:0	R/W
0x8	Ch A counter wrap		
	<i>Reserved</i>	31:30	R
	Value where counter wraps around. Depends on SM wrap setting. If it is 1 new value is get by wrap, if value is 0 counter goes to offset value. 16 bits for decimals.	29:0	R/W
0xC	Ch A start offset		
	<i>Reserved</i>	31:30	R
	Counter start offset. Start offset when trigger arrives. 16 bits for decimals.	29:0	R/W

0x10	Ch A counter step		
	<i>Reserved</i>	<i>31:30</i>	<i>R</i>
	Counter step. 16 bits for decimals.	29:0	R/W
0x14	Ch A buffer current read pointer		
	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Read pointer	15:2	R/W
	<i>Reserved</i>	<i>1:0</i>	<i>R</i>
0x18	Ch A number of read cycles in one burst		
	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Number of repeats of table readout. 0=infinite	15:0	R/W
0x1C	Ch A number of burst repetitions		
	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Number of repetitions. 0=disabled	15:0	R/W
0x20	Ch A delay between burst repetitions		
	Delay between repetitions. Granularity=1us	31:0	R/W
0x24	Ch B amplitude scale and offset		
	out = (data*scale)/0x2000 + offset		
	<i>Reserved</i>	<i>31:30</i>	<i>R</i>
	Amplitude offset	29:16	R/W
	<i>Reserved</i>	<i>15:14</i>	<i>R</i>
	Amplitude scale. 0x2000 == multiply by 1. Unsigned	13:0	R/W
0x28	Ch B counter wrap		
	<i>Reserved</i>	<i>31:30</i>	<i>R</i>
	Value where counter wraps around. Depends on SM wrap setting. If it is 1 new value is get by wrap, if value is 0 counter goes to offset value. 16 bits for decimals.	29:0	R/W
0x2C	Ch B start offset		
	<i>Reserved</i>	<i>31:30</i>	<i>R</i>
	Counter start offset. Start offset when trigger arrives. 16 bits for decimals.	29:0	R/W
0x30	Ch B counter step		
	<i>Reserved</i>	<i>31:30</i>	<i>R</i>
	Counter step. 16 bits for decimals.	29:0	R/W
0x34	Ch B buffer current read pointer		
	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Read pointer	15:2	R/W
	<i>Reserved</i>	<i>1:0</i>	<i>R</i>
0x38	Ch B number of read cycles in one burst		

	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Number of repeats of table readout. 0=infinite	15:0	R/W
0x3C	Ch B number of burst repetitions		
	<i>Reserved</i>	<i>31:16</i>	<i>R</i>
	Number of repetitions. 0=disabled	15:0	R/W
0x40	Ch B delay between burst repetitions		
	Delay between repetitions. Granularity=1us	31:0	R/W
0x10000 to 0x1FFFC	Ch A memory data (16k samples)		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	ch A data	13:0	R/W
0x20000 to 0x2FFFC	Ch B memory data (16k samples)		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	ch B data	13:0	R/W

PID Controller

offset	description	bits	R/W
0x0	Configuration		
	<i>Reserved</i>	<i>31:4</i>	<i>R</i>
	PID22 integrator reset	3	R/W
	PID21 integrator reset	2	R/W
	PID12 integrator reset	1	R/W
	PID11 integrator reset	0	R/W
0x10	PID11 set point		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID11 set point	13:0	R/W
0x14	PID11 proportional coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID11 Kp	13:0	R/W
0x18	PID11 integral coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID11 Ki	13:0	R/W
0x1C	PID11 derivative coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID11 Kd	13:0	R/W
0x20	PID12 set point		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID12 set point	13:0	R/W
0x24	PID12 proportional coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID12 Kp	13:0	R/W
0x28	PID12 integral coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID12 Ki	13:0	R/W
0x2C	PID12 derivative coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID12 Kd	13:0	R/W
0x30	PID21 set point		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID21 set point	13:0	R/W

0x34	PID21 proportional coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID21 Kp	13:0	R/W
0x38	PID21 integral coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID21 Ki	13:0	R/W
0x3C	PID21 derivative coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID21 Kd	13:0	R/W
0x40	PID22 set point		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID22 set point	13:0	R/W
0x44	PID22 proportional coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID22 Kp	13:0	R/W
0x48	PID22 integral coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID22 Ki	13:0	R/W
0x4C	PID22 derivative coefficient		
	<i>Reserved</i>	<i>31:14</i>	<i>R</i>
	PID22 Kd	13:0	R/W

Analog Mixed Signals (AMS)

offset	description	bits	R/W
0x0	XADC AIF0		
	<i>Reserved</i>	<i>31:12</i>	<i>R</i>
	AIF0 value	11:0	R
0x4	XADC AIF1		
	<i>Reserved</i>	<i>31:12</i>	<i>R</i>
	AIF1 value	11:0	R
0x8	XADC AIF2		
	<i>Reserved</i>	<i>31:12</i>	<i>R</i>
	AIF2 value	11:0	R
0xC	XADC AIF3		
	<i>Reserved</i>	<i>31:12</i>	<i>R</i>
	AIF3 value	11:0	R
0x10	XADC AIF4		
	<i>Reserved</i>	<i>31:12</i>	<i>R</i>
	AIF4 value (5V power supply)	11:0	R
0x20	PWM DAC0		
	<i>Reserved</i>	<i>31:24</i>	<i>R</i>
	PWM value (100% == 156)	23:16	R/W
	Bit select for PWM repetition which have value PWM+1	15:0	R/W
0x24	PWM DAC1		
	<i>Reserved</i>	<i>31:24</i>	<i>R</i>
	PWM value (100% == 156)	23:16	R/W
	Bit select for PWM repetition which have value PWM+1	15:0	R/W
0x28	PWM DAC2		
	<i>Reserved</i>	<i>31:24</i>	<i>R</i>
	PWM value (100% == 156)	23:16	R/W
	Bit select for PWM repetition which have value PWM+1	15:0	R/W
0x2C	PWM DAC3		
	<i>Reserved</i>	<i>31:24</i>	<i>R</i>
	PWM value (100% == 156)	23:16	R/W
	Bit select for PWM repetition which have value PWM+1	15:0	R/W

Daisy Chain

offset	description	bits	R/W
0x0	Control		
	<i>Reserved</i>	<i>31:2</i>	<i>R</i>
	RX enable	1	R/W
	TX enable	0	R/W
0x4	Transmitter data selector		
	Custom data	31:16	R/W
	<i>Reserved</i>	<i>15:8</i>	<i>R</i>
	Data source 0-data is 0 1-user data (from logic) 2-custom data (from this register) 3-training data (0x00FF) 4-transmit received data (loop back) 5-random data (for testing)	3:0	R/W
0x8	Receiver training		
	<i>Reserved</i>	<i>31:2</i>	<i>R</i>
	Training successful	1	R
	Enable training	0	R/W
0xC	Received data		
	Received data which is different than 0	31:16	R
	Received raw data	15:0	R
0x10	Testing control		
	<i>Reserved</i>	<i>31:1</i>	<i>R</i>
	Reset testing counters (error & data)	0	R/W
0x14	Testing error counter		
	Error increases if received data is not the same as transmitted testing data	31:0	R
0x18	Testing data counter		
	Counter increases when value different as 0 is received	31:0	R

Power Test

offset	description	bits	R/W
0x0	Control		
	<i>Reserved</i>	<i>31:1</i>	<i>R</i>
	Enable module	0	R/W